

A1. Destroy all monsters

You are playing a video game in which there are **N** number of monsters present and each monster has a specific power which is stored in integer array **A**. Now your goal is to defeat all the monsters

Guidelines

1. Only two monsters can clash at the same time. After the clash, the energy cost is equal to the GCD of their powers. It can only clash whose position is the start, middle, or end of the array

2. Your task is to find and return an integer value denoting the minimum total energy required to kill all monsters

Input1:

An integer **N** represents the number of elements in an array

Array **a** denoting the power of each monster

Output

Return an integer denoting the minimum total energy required to kill monster

Input1

4
1234

Output

2

A2. MAX OR ARRAY

Jonathan is utilizing his summer vacations learning about arrays. He is solving a problem and gets stuck at some point so he asks you for help.

You are given an array **A** of size **N** containing non-negative integers. You can remove a subarray from array **A** such that the bitwise **OR** of all the elements in the remaining array is the maximum possible.

Your task is to find and return an integer value the size of the maximum length subarray that can be removed.

Input Specification:

input1: An integer array **A**

input2: An integer **N** denoting the size of the array

Output Specification:

Return an integer value the size of the maximum length subarray that can be removed.

Example 1

input1: {1,4,24,2}

input2: 4

Output: 0

Explanation:

The bitwise **OR** of the whole array is **31** which is the maximum value possible and it cannot be achieved by removing any subarray. Hence, **0** is returned as the output.

A3. Longest Valid Path

You are given 2D integer of size N/M with distinct elements.

From any cell you can move to any adjacent cell if and only if the element in the adjacent cell is less than the element in the current cell and the GCD of both the elements is greater than 1

Your task is to find an integer value representing the length of the longest path that starts and ends at any cell in the 2D array.

Input1:

An integer value N – NO OF rows of 2D array

Input2:

M- NO OF columns of the 2D array

Input:

3

2

Output:

1({1 2} {5 6} {4 9})

A4. Sub number greater than K

A sub number of a given number is a value that can be derived from the given number by deleting sum or no digits without changing the order of remaining digits.

Given 2 non-negative integers N and K you task is to return the sub number of N which is greater than K and also closed to K.

Note: K is always less than N

Input1:

An integer value denoting the non- negative number integer N

Input2:

An integer value denoting the non -negative number integer K

Output:

Return an integer representing the sub number of N

Input1:

1 2 3
1 4

Output

2 3

B1. DECAY TRAIL OPTIMIZATION

You are working with a climate research lab that tracks the decay of biomass on a 2D terrain. The terrain is represented as a matrix of cells, where each cell contains an integer value representing the biomass concentration at that location.

A researcher wants to analyse how quickly the decay can propagate recursively where each propagation can only move to an adjacent cell from a specific cell.

From any cell, the decay can propagate to an adjacent cell only if the adjacent cell has a strictly lower biomass concentration than the current cell.

Movement is allowed only in the following four directions:

Up

Down

Left

Right

Diagonal movement is not allowed.

Given the biomass matrix and a starting cell, determine the maximum possible length of a strictly decreasing decay trail that can be formed starting from that cell.

Input:

The input consists of:

Two integers N and M, representing the number of rows and columns in the matrix.

An $N \times M$ matrix of integers representing biomass concentrations.

Two integers X and Y, representing the 0-based row and column indices of the starting cell.

Constraints

$1 \leq N, M$

$1 \leq N \times M$ within the specified problem limits

Biomass values are integers.

$0 \leq X < N$

$0 \leq Y < M$

Output

Print a single integer representing the maximum length of the strictly decreasing decay trail starting from cell (X, Y).

The starting cell itself is counted as part of the trail.

Example

Input

3 3
9 6 3
8 5 2
7 4 1
0 0

Explanation

Starting from (0,0) with biomass value 9, one possible decreasing trail is: $9 \rightarrow 8 \rightarrow 7 \rightarrow 4 \rightarrow 1$

The trail has a length of 5.

Therefore, the output is: Output 5

Notes

A move is valid only when the destination cell contains a strictly smaller value than the current cell.

Cells with equal biomass values cannot be visited consecutively.

The path cannot move outside the matrix boundaries.

Since every move strictly decreases the biomass value, the trail cannot contain a cycle.

Objective Determine the maximum possible strictly decreasing decay-trail length starting from the given cell (X, Y).

B2. Maximum reward points

Alex is given a path containing N nodes. Each node contains an associated reward value.

He wants to maximize the sum of reward points collected from non-adjacent nodes along this path.

However, there is a unique rule: Alex can select a node and its immediate predecessor in the path is negative.

If the predecessor has a negative value, he must include both nodes in the sum and subtract the value of the negative node.

Your task is to help Alex find and return an integer representing the maximum sum of reward points that can be collected.

Input

N — number of nodes.

An array of N integers representing the reward value of each node.

Example

Input:

N = 7

A = [1, 4, 8, -3, 10, 6, 1]

Output:

13

B3. PRIME TRANSFORMATION

In a secret cryptography system — every number must transform into a prime code based on a special digit transformation rule.

You are given a +ve Int N and you can transform to obtain a prime using the following operation:

→ In each step, you can choose one digit & increase or decrease it by 1 (for e.g. 433 or 475)

→ You can perform at most K such digit operations across the number.

After at most K changes you must check whether any transformed version of N becomes a prime number.

Your task is to find and return an integer value representing the smallest possible prime number obtainable from N using at most K single digit (+1) changes.

If no such transformation leads to a prime number, return -1

Input 1:

int N — you must transform into a prime number. 2:

int K — denotes the number of digit operations you can perform.

Output

Return an int value to represent the smallest possible prime number obtainable from N using at most K single digit (± 1) changes. If no such transformation leads to a prime number, return -1.

Example

Input:

231

2

Output:

223

B4. Minimum Moves

Minimum Moves to Reach the Hospital A patient is in a city under quarantine due to a virus outbreak. The city is represented by N checkpoints arranged in a linear sequence. Each checkpoint has a non-zero integer value that indicates the distance the patient can move from that checkpoint.

If the patient is currently at checkpoint i, they can make a move to either: $i + A[i]$ (move $A[i]$ steps to the right), or $i - A[i]$ (move $A[i]$ steps to the left). A move cannot be made if it takes the patient outside the range of the available checkpoints.

There is exactly one checkpoint containing a hospital that provides a life-saving vaccine. This hospital checkpoint is represented by the value 0.

Given the starting checkpoint, find and return the minimum number of moves required for the patient to reach the hospital.

If it is not possible for the patient to reach the hospital, return -1.

Input

N — an integer representing the number of checkpoints.

A — an integer array of size N, where $A[i]$ represents the number of positions the patient can move from checkpoint i. Exactly one element in the array is 0, representing the hospital.

Output

Return an integer representing the minimum number of moves required to reach the hospital from the given starting checkpoint. If the hospital cannot be reached, return -1.

B5. Lego String / Magic Lego Chain

You are given a **wall string** and a set of **LEGO blocks**. Your task is to find all starting positions in the wall where a **magic LEGO chain** can be formed.

A magic LEGO chain is a **contiguous substring** of the wall where **all the given LEGO blocks appear exactly once, in any order**, joined together **without any spaces or gaps**.

Each block can be used **at most once**.

Task

Given the wall string and an array of LEGO blocks, find all starting indices (0-based) where a valid LEGO chain can be formed.

If no such position exists, return -1.

Input

1. A string wall, representing the wall text.
2. An integer N, representing the number of LEGO blocks.
3. An array of N strings B[], representing the LEGO blocks.

Output

Return an integer array containing all the **starting positions (0-based)** where a valid LEGO chain can be formed.

If no valid position exists, return [-1].

Example

Suppose:

wall = "barfoothefoobarman"

N = 2

B = ["foo", "bar"]

The valid LEGO chains are:

"barfoo" → starting index 0

"foobar" → starting index 9

Therefore:

Output: [0, 9]

The key idea is that the blocks may appear in **any order**, but they must be **adjacent with no spaces or extra characters**.

